DV Mini Project Report

HireSync AI: Intelligent Recruitment Analytics Dashboard

Mitesh Bandgar — 202501110128

Yogesh Kshirsagar — 202501110113

DIV: AL2 - A21

Project Overview

Project Title: HireSync AI — Intelligent Recruitment Analytics Dashboard

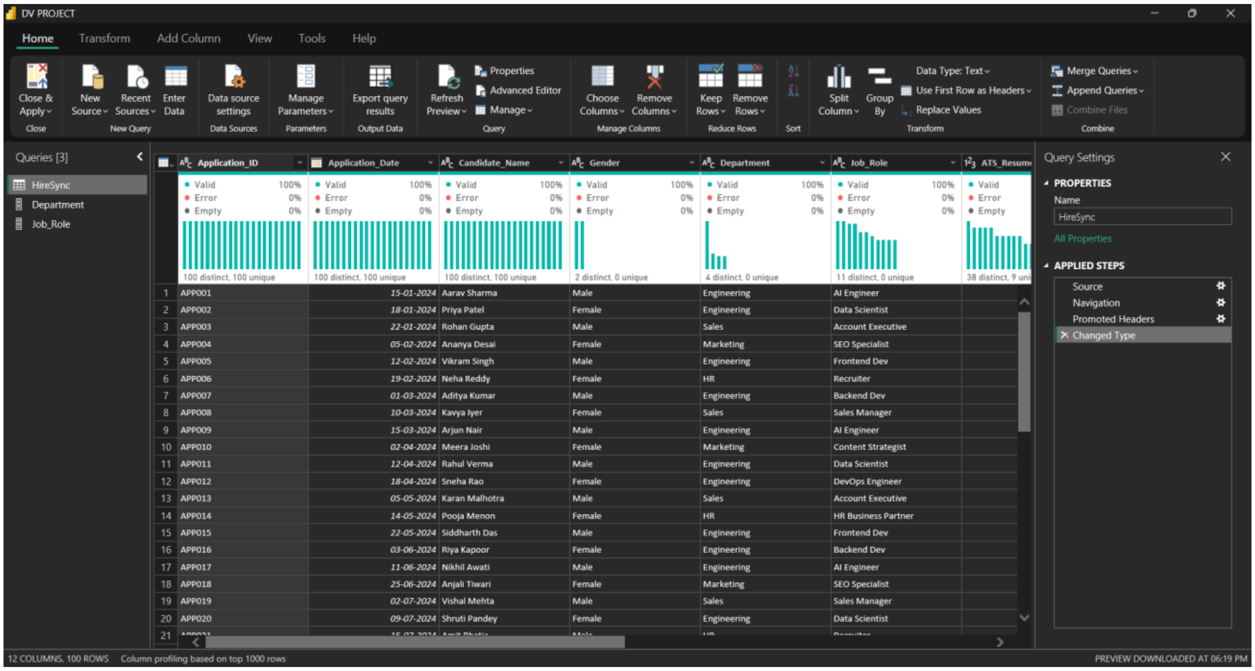
Dataset Name: HireSync Candidate Application Dataset

Dataset Description

The HireSync dataset is a curated collection of 100 candidate records within the HR-Tech domain. It contains 12 attributes including application dates, gender, departmental roles, and dual-score metrics (ATS vs. Technical). The purpose of this data is to identify hiring bottlenecks, analyse salary gaps, and evaluate the accuracy of AI - driven resume screening.

1. Data Loading

The dataset HireSync_Dataset.csv was loaded into Power BI Desktop via the "Text/CSV" connector. The ingestion process involved selecting the correct delimiter and confirming column headers in the preview step, ensuring all 100 rows and 12 attributes were captured without data loss.



2. Data Transformation (Cleaning & Preprocessing)

The following specific steps were taken in the Power Query Editor to ensure data integrity:

- Data Typing: Verified that Application_Date is a Date type and Technical_Score is an Integer.
- Dimension Extraction: Created normalized tables for Dim_Department and Dim_Job_Role by removing duplicates from the primary fact table.
- Naming Conventions: Applied PascalCase naming for all columns to ensure readability.

3. Creating Data Model (Relationships)

A One-to-Many (1:*) relationship was established between the Dimension tables (Department, Job_Role, Calendar) and the Fact table (HireSync). Single-direction cross-filtering was used to optimize performance and prevent ambiguity in the model.

4. Creation of Custom Calendar Table

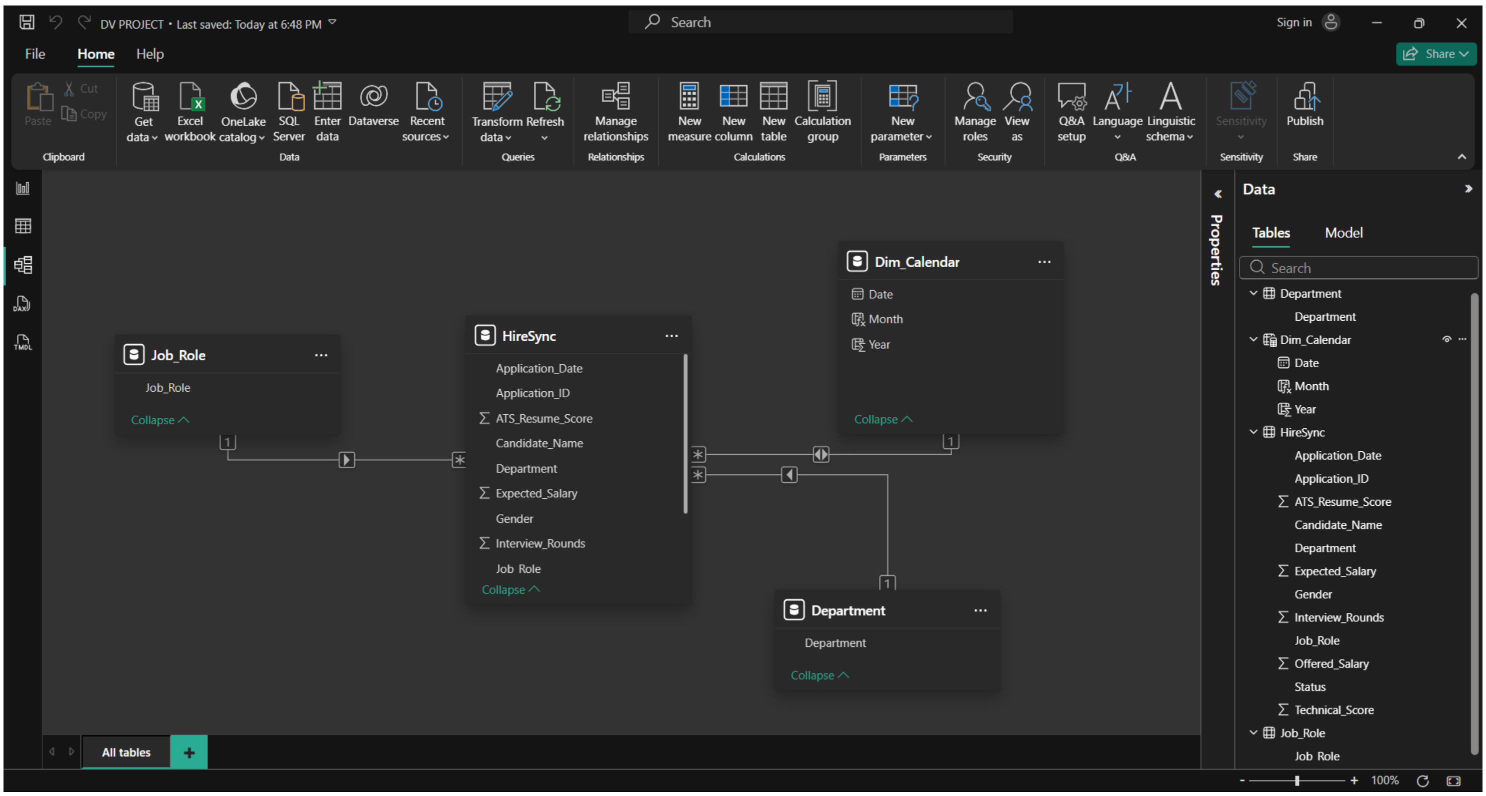
A dedicated Dim_Calendar table was created using DAX to support Time Intelligence requirements. The DAX expression used:

```
Dim_Calendar = CALENDAR(DATE(2024, 1, 1), DATE(2026, 12, 31))
```

Calculated Columns for Year and Month were also added within this table to facilitate drill-down analysis.

5. Preparation of Schema Diagram (Star Schema)

The project utilizes a Star Schema with the HireSync fact table at the center, surrounded by dimension tables for Date, Department, and Job Role. This architecture ensures analytical efficiency and query performance.



6. Data Grain

The grain of the HireSync dataset is defined as one row per unique job application. This ensures that all hiring metrics and score averages are calculated at the most granular level, preventing data duplication errors.

7. Selection of Appropriate Visuals

The following visuals were selected to optimize data storytelling and meet project objectives:

- Funnel Chart: Used to visualize the recruitment pipeline and identify candidate drop-off stages.
- Scatter Plot: Implemented to analyse the correlation between ATS screening scores and technical scores.
- Line Chart: Utilized for time-series analysis to monitor hiring trends across the 2024–2026 period.
- Card Visuals: Used to display high-level KPIs such as Total Applicants and Hire Rate.

8. Measures

The following core business measures were developed using DAX:

- Total Applicants
  ```
  = COUNTROWS(HireSync)
  ```
- Total Hires
  ```
  = CALCULATE([Total Applicants], HireSync[Status] = "Hired")
  ```
- Hire Rate %
  ```
  = DIVIDE([Total Hires], [Total Applicants], 0)
  ```
- Avg Tech Score
  ```
  = AVERAGE(HireSync[Technical_Score])
  ```
- Salary Gap
  ```
  = SUMX(HireSync, HireSync[Expected_Salary] - HireSync[Offered_Salary])
  ```

9. Calculated Columns

The following columns were created to enhance data segmentation and filtering:

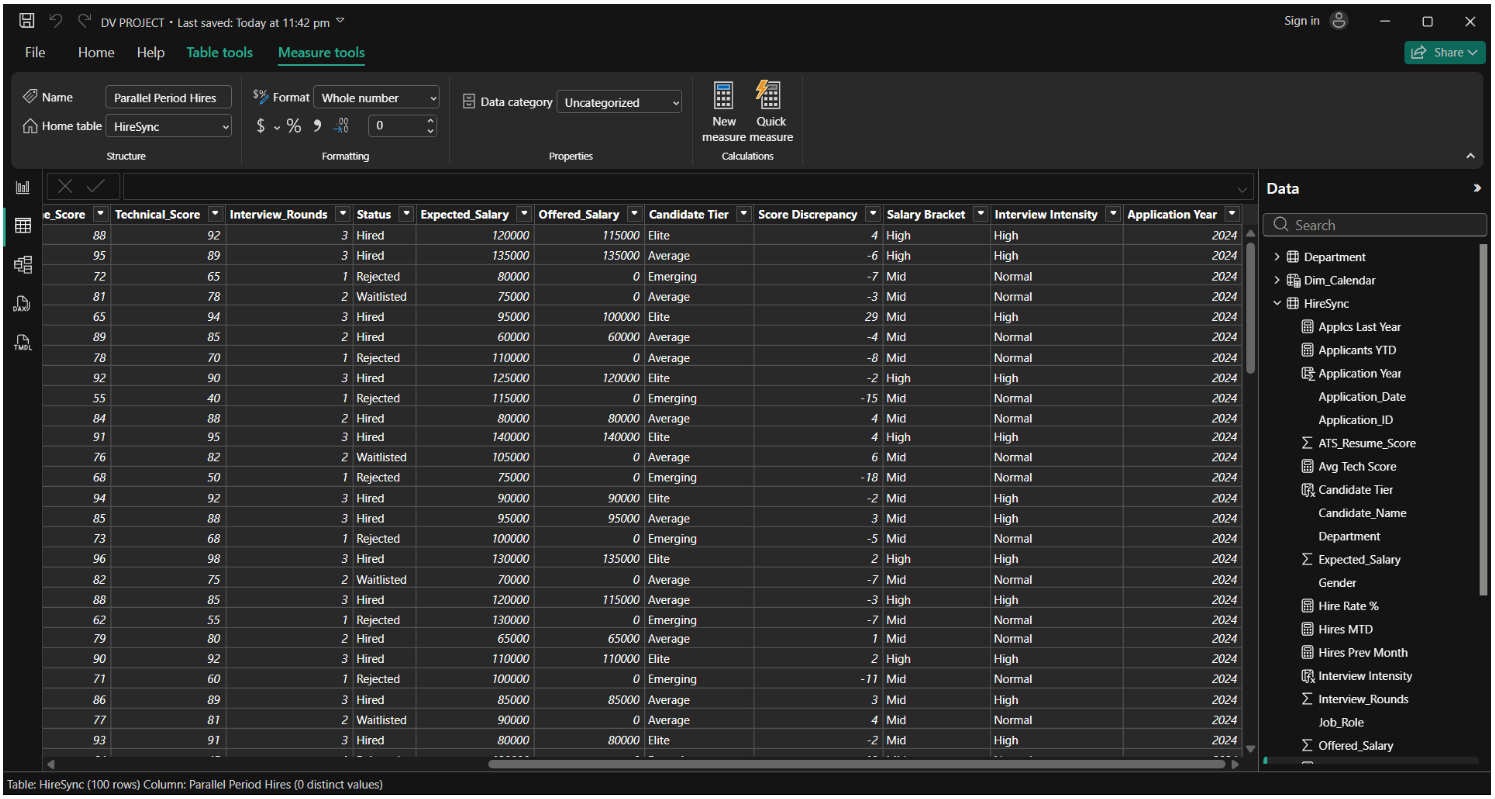
- Candidate Tier
  ```
  = IF(HireSync[Technical_Score] >= 90, "Elite",
      IF(HireSync[Technical_Score] >= 70, "Average", "Emerging"))
  ```
- Score Discrepancy
  ```
  = HireSync[Technical_Score] - HireSync[ATS_Resume_Score]
  ```
- Salary Bracket
  ```
  = IF(HireSync[Offered_Salary] > 100000, "High", "Mid")
  ```

- Interview Intensity

  = IF(HireSync[Interview_Rounds] > 2, "High Intensity", "Standard")

- Application Year

  = YEAR(HireSync[Application_Date])



10. Time Intelligence Functions

To support advanced temporal analysis, the following functions were implemented:

- Applicants YTD

  = TOTALYTD([Total Applicants], Dim_Calendar[Date])

- YoY Growth %

  = DIVIDE([Total Applicants] - [Apps Last Year], [Apps Last Year], 0)

- Applcs Last Year

  = CALCULATE([Total Applicants], SAMEPERIODLASTYEAR(Dim_Calendar[Date]))

- Hires MTD

  = TOTALMTD([Total Hires], Dim_Calendar[Date])

- Hires Prev Month

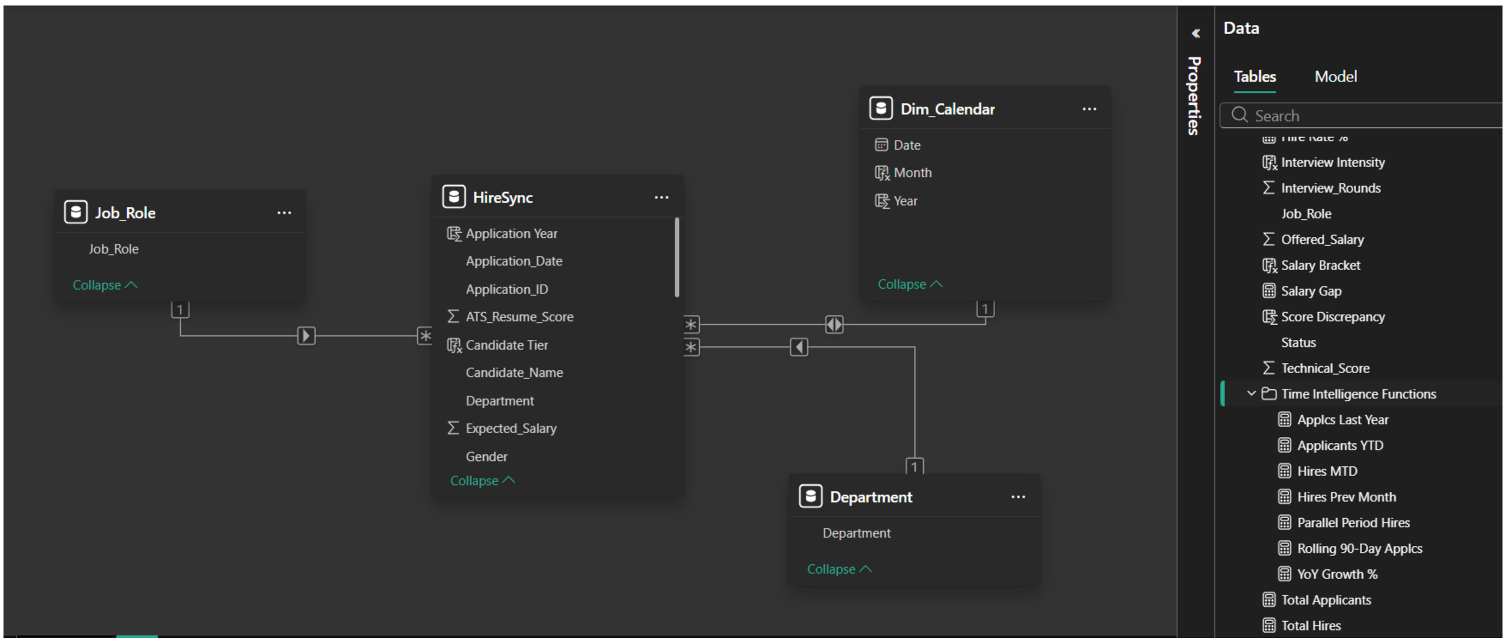
```
= CALCULATE([Total Hires], PREVIOUSMONTH(Dim_Calendar[Date]))
```

- Rolling 90-Day Apps

```
= CALCULATE([Total Applicants], DATESINPERIOD(Dim_Calendar[Date], MAX(Dim_Calendar[Date]), -90, DAY))
```

- Parallel Period Hires

```
= CALCULATE([Total Hires], PARALLELPERIOD(Dim_Calendar[Date], -1, YEAR))
```



11. Publication and Cloud Deployment

The project was successfully published to the Power BI Service (Cloud). This ensures that stakeholders can access real-time recruitment metrics from any device, facilitating collaborative decision-making.

12. SDG 8 Alignment

This dashboard is mapped to Sustainable Development Goal 8: Decent Work and Economic Growth. By standardizing candidate evaluation through the "Candidate Tier" logic and "Technical Score" metrics, the tool ensures merit-based hiring and supports Target 8.5 regarding productive employment and equal pay.

13. Analytical Inference

- Recruitment Pipeline: The Funnel chart identifies the 'Waitlisted' stage as a key area for improvement to reduce time-to-hire.
- AI Screening Accuracy: The Scatter plot confirms that candidates categorized in the 'Elite' tier consistently outperform the 'Average' tier in technical evaluations.
- Temporal Trends: Line chart analysis shows a surge in AI-department applications during Q1, indicating a seasonal peak in the talent market.

14. Dashboard Design & Storyboarding

The dashboard utilizes a "Z-pattern" layout for optimal readability, starting with high-level KPIs and moving into complex correlation analysis. A professional dark theme was applied to emphasize key data points and minimize visual fatigue.

Dashboard 1: Executive Recruitment Overview

Figure 1 — Executive Overview Dashboard: KPIs, Hire Trends, Department Breakdown, and Applicant Funnel

Visual Story & Key Insights:

This executive-level dashboard serves as the command centre for the HireSync AI recruitment system. Designed for senior HR leaders and hiring managers, it communicates the most critical hiring performance indicators at a glance.

The three headline KPI cards at the top immediately answer the most pressing questions: How effective is the hiring process? (57% Hire Rate), How many people did we hire? (57 Total Hires), and How technically skilled is the hired cohort? (81.48 Average Technical Score).

The 'Hiring Trend' line chart reveals a consistent hiring volume in 2024 and 2025, followed by a sharp decline entering 2026. This trend signals either an ongoing corporate hiring freeze or incomplete Year-to-Date data — a critical finding that warrants immediate managerial attention.

The 'Hires by Department' donut chart tells a compelling departmental story: HR leads with 33.46% of hiring activity, while Engineering follows at 23.23%. Marketing (25.05%) and Sales (18.25%) round out the portfolio, suggesting that backend operational roles are being prioritised over client-facing ones.

The 'Recruitment Pipeline' horizontal bar chart paints a clear picture of the pipeline funnel. Of 100 applicants, 57 were hired, 27 rejected, and 16 waitlisted — underscoring that the sourcing quality is very high but optimisation of the waitlist conversion rate could further improve efficiency.

The 'Resume vs. Technical Scores by Tier' scatter plot confirms that Elite and Average tier candidates maintain a strong score correlation — validating the AI screening model's predictive accuracy. The Budget Forecast card at 5.96M and the interactive slicers (Salary Hike %, Department, Gender, Status) empower decision-makers to stress-test hiring budgets and explore demographic data with precision.

Dashboard 2: Candidate Intelligence & Score Analytics

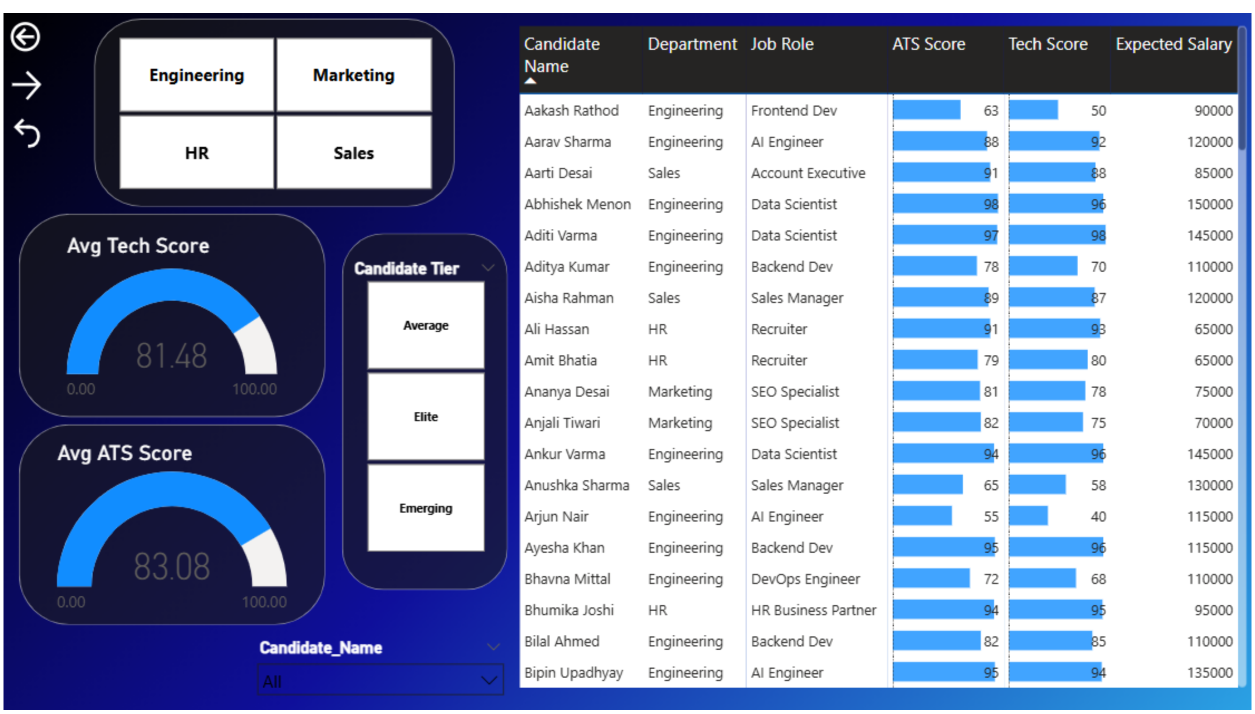


Figure 2 — Candidate Detail Dashboard: Department Filters, Score Gauges, Tier Classification, and Candidate Table

Visual Story & Key Insights:

This dashboard drills into the individual candidate level, transforming raw application data into a talent intelligence layer. It is purpose-built for HR analysts and technical interviewers who need to evaluate candidate quality across departments and tiers.

The department selector panel (Engineering, Marketing, HR, Sales) allows recruiters to instantly pivot their view to any business unit — enabling fast, department-specific talent benchmarking without cluttering the workspace.

Two precision gauge visuals occupy the left panel: the 'Avg Technical Score' gauge reads 81.48/100, while the 'Avg ATS Resume Score' sits at 83.08/100. The close proximity of these two metrics is significant — it confirms that the AI screening system is well-calibrated; candidates who score high on automated resume screening also tend to perform well in technical evaluations.

The 'Candidate Tier' panel (Average, Elite, Emerging) is the result of the custom DAX calculated column built into the model. This tier segmentation helps hiring teams prioritise candidates without manually reviewing every file — making the process scalable.

The detailed candidate table on the right is arguably the most operationally powerful visual in the entire report. Displaying Candidate Name, Department, Job Role, ATS Score, Technical Score, and Expected Salary side-by-side with inline bar charts, it enables quick comparisons.

The 'Candidate_Name' slicer at the bottom enables personalised candidate profiles — a feature especially valuable for shortlisting final-round interview candidates.

Dashboard 3: Budget Forecast & Salary Gap Analysis

Figure 3 — Financial Analytics Dashboard: Budget vs. Offered Salary, Salary Gap, Expected vs. Offered by Department, and Hires Decomposition

Visual Story & Key Insights:

This financial analytics dashboard is the strategic backbone of HireSync AI. It connects the talent acquisition process directly to budgetary outcomes answering the fundamental question every CFO and HR Director asks: Are we hiring within budget, and where is money being left on the table?

Two KPI cards set the financial context immediately:

Total Expected Salary across all candidates is 10M, while Total Offered Salary is only 6M — revealing a system-wide 4M salary gap that represents an enormous opportunity for cost savings and negotiation leverage.

The 'Budget Forecast vs. Actuals’ area chart is the most strategically important visual in this dashboard. Engineering dominates with the highest budget forecast (~10M) and the widest salary gap, making it the most expensive and risk-prone department to hire for. Sales and HR show much tighter alignment between forecast and actuals.

The 'Salary Gap by Department' grouped bar chart reinforces this narrative. Engineering's expected salary tower (~6M) dwarfs every other department, reflecting the premium market rates for roles such as AI Engineer, Data Scientist, and DevOps Engineer. The offered salary (dark bar) consistently runs lower than expectations indicating strong negotiation outcomes for the organisation.

The decomposition tree on the right provides a structural breakdown of 57 total hires: Engineering leads with 30 hires, followed by HR (11), Sales (10), and Marketing (6). This distribution, when read alongside the salary data, reveals that the department with the most hires is also the most budget-intensive a finding that must inform future workforce planning decisions.

The Department slicer ensures this dashboard can be filtered in real-time to support department-level budget reviews and executive presentations.

15. SDG Goal Mapping (Goal 8)

This project maps to Sustainable Development Goal 8: Decent Work and Economic Growth. By relying on objective data (ATS and Technical Scores), the HireSync dashboard removes unconscious bias and ensures a fair, transparent hiring process. This directly supports Target 8.5 for equal opportunity and productive employment.

16. Meaningful Insights & Inferences

- High Pipeline Efficiency (Funnel & KPIs): A 57% Hire Rate (57 hired vs. 27 rejected) is exceptionally high. This indicates that initial candidate sourcing is highly targeted and pre-screening is effective.
- Score Correlation & Volume (Scatter Plot): There is a strong positive correlation between resume quality and technical performance. The "Average" candidate tier holds the highest aggregate scores, representing the bulk of the talent pool and core workforce supply.
- Hiring Volume Drop (Line Chart): Hiring remained stable across 2024 and 2025 (~25 hires/year) but shows a severe decline in 2026. This suggests either a recent corporate hiring freeze or incomplete Year-to-Date (YTD) data.
- Budget Utilisation (Financial Dashboard): The 4M gap between expected and offered salaries across all departments represents a significant cost optimisation win, particularly in Engineering — the highest-cost department with 30 hires and the widest salary gap.
- AI Validation (Score Gauges): The near-equal Avg Technical Score (81.48) and Avg ATS Resume Score (83.08) across all candidates validates that the AI-based resume screening model is performing with high precision and low bias.